

Krishnakumar Ramesan

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Senior Software Engineer | Distributed Systems | Multi-Agent AI | SSO & Event-Driven Architecture

PROFILE

Senior Software Engineer with **13+ years** building enterprise microservices in **Kotlin/Java**, identity integrations on **Okta** using **SAML and OIDC**, and event-driven systems on **Kafka and RabbitMQ**. Designed and built a production multi-agent AI onboarding assistant for QuickBooks on **LangGraph and OpenAI GPT-4.1**, shipped to new trial customers. Comfortable owning ambiguous, high-stakes systems end to end, from design through GA and production hardening.

EXPERIENCE

Intuit - Senior Software Engineer

Aug 2022 - Present

- **Designed and shipped** a production multi-agent AI onboarding assistant for QuickBooks on LangGraph and OpenAI GPT-4.1 as tech lead, taking it from hackathon prototype to a rollout used by new trial customers. Built a config-driven intent orchestrator that classifies each message and routes across 7 domain agents.
- **Built and hardened** the screen context engine that grounds those agents in what the customer sees, converting the live DOM into compact markdown with on-page redaction of sensitive data like SSNs, addresses, and phone numbers. Replaced a multi-second screenshot-plus-vision-LLM pipeline with a near-instant DOM parse, debounced and navigation-aware to stay accurate without flooding the pipeline.
- **Designed and built** account and credential management for QuickBooks' shipping module, launched to GA as the **first core revenue product in the Commerce Business Unit**. Secured thousands of carrier API credentials with AES-256-GCM envelope encryption via AWS KMS (30-day key rotation, mTLS), built as an isolated migration-ready module to unblock GA when the platform could not support the provider's hierarchical key model.
- **Owned** a fault-tolerant fulfillment subsystem ingesting events from **4 sales channels** (hundreds of thousands of events per day), designed for idempotent consumers, ordered delivery, and safe recovery under partial failures. Fulfillment events are linked to upstream product and variant entities before being published for downstream inventory consumers, making ordering a core design constraint. Scaled the pipeline to sustain **75 TPS** through database partitioning, sharding, and load-based autoscaling, verifying throughput with custom load-generation scripts.
- **Diagnosed and optimized** a bulk-match API failing under dense product/variant relationships, removing N+1 round-trips and over-fetching and adding targeted indexes to **cut response time from 30 s to 785 ms (97%)**.
- **Designed and built** a configurable peer-review system for Live Bookkeeping expert work with FIFO-based reviewer assignment, structured questionnaires with bidirectional notifications, and UiPath RPA integration for automated final-stage review. New services onboard through configuration alone, **saving around 25K hours of expert time across 50K active services**.

TriNet - Software Engineer

Aug 2016 - Jul 2022

- **Built** a scalable ACL-based authorization service consumed by **20 microservices**, eliminating per-request database round-trips with a two-tier cache (per-user role entries invalidated on role-change events, endpoint access policies at a longer TTL), measurably reducing p90 authorization latency under production load. Delivered a companion dynamic policy dashboard allowing DevOps to update role requirements without code redeploys across all 20 dependent services.
- **Built** a core SSO microservice serving **~400K TriNet customers** across **~20 cloud software provider integrations**, resolving provider-specific SAML attributes, surfacing per-user provisioned application links, and orchestrating the final SSO handshake to enable password-free access for employees with no dedicated IT resources.
- **Eliminated** dependency on live partner environments for SSO regression testing by building a SAML integration test stub covering **~20 provider integrations**, enabling QE to run full regression suites on-demand and decoupling release cycles from third-party availability.

- **Built** an audited event-bridge connecting an on-premises HR engine to new microservices, processing **300-400K events/day** with write-ahead audit logging as a replay buffer for failure recovery. Delivered an admin dashboard for support to query, correct, and resubmit failed events, **saving 10-15 min per incident**, with a full change audit trail on every human intervention.

Mazda North American Operations - Contractor

Nov 2015 - Apr 2016

- Built RESTful microservices during a monolith modernization initiative, delivering performance improvements and feature enhancements to internal applications.

Infosys Limited - Systems Engineer

Jun 2011 - Jun 2013

- Monitored and supported production ETL jobs, migrated and tested code across lower environments, and reported daily job statistics to business stakeholders.

EDUCATION

M.S. Computer Science, University of Alabama in Huntsville - May 2015 | GPA **3.75 / 4.0**

B.Tech Electronics and Communication, Amrita School of Engineering - Jun 2011 | GPA **3.0 / 4.0**

SKILLS

Languages: Kotlin, Java, Python

Frameworks: Spring Boot, Hibernate, GraphQL (Netflix DGS)

AI / Agents: LangGraph, OpenAI API (GPT-4.1), multi-agent orchestration, DOM-based context engineering

Messaging / Streaming: Kafka, RabbitMQ

Data: PostgreSQL, DynamoDB, Redis

Identity / Security: Okta, SAML, OIDC, SSO, AWS KMS

Platform: AWS, Kubernetes, Podman